Summer Project

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Abstract

This project studies how to keep an optical system working properly when a thermo-optical medium slowly drifts over time. A Gaussian beam is first phase-shaped by a spatial light modulator (SLM), then passes through a medium whose optical response changes as its temperature evolves, and is finally measured by a detector through its output intensity distribution. The central goal is to maintain a desired output intensity pattern at the output plane even as thermal drift changes the refractive index of the medium, alters the optical phase accumulated by the beam, and reshapes the interference pattern seen at the detector.

The project is organized around two connected tasks. The first is to model the evolving thermal state of the medium and its effect on the optical response, using either a physics-informed neural network (PINN) or a classical numerical solution of the heat equation. The second is to use feedback from the measured output intensity to update the SLM phase pattern so that the desired output intensity pattern can be maintained or recovered over time. This feedback loop may be implemented using reinforcement learning, with the reward based on how well the measured output matches the desired response.

In this way, the project separates the problem into modelling and control: the thermal model describes how and why the system drifts, while the controller decides how the wavefront should be adjusted to compensate for that drift and preserve the target intensity distribution.

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1 Experimental Setup

2 Experimental Setup

This project studies a closed-loop optical system whose output degrades because the medium through which the beam propagates is thermally drifting. A Gaussian beam is first phase-shaped by a spatial light modulator (SLM), then sent through a thermo-optical material, and finally measured at a detector. The main objective is to maintain a desired output intensity pattern at the output plane, even though the optical response of the medium changes over time.

The physical mechanism behind this drift is as follows. As the temperature field inside the medium evolves, the refractive index changes through the thermo-optic effect. These refractive-index changes alter the optical path length and therefore the phase accumulated by the beam during propagation. Because the measured output is formed by interference of many field contributions, phase drift in the medium leads directly to drift in the output intensity distribution. In compact form,

$$\text{thermal drift} \longrightarrow \Delta n(\mathbf{r}, t) \longrightarrow \Delta\phi(x, y, t) \longrightarrow \text{interference change} \longrightarrow I(x, y) \text{ drift.}$$

An SLM phase pattern that produces the desired output at one moment will therefore generally stop doing so as the medium continues to evolve.

The project is organized around four coupled components. First, a thermal model describes the evolving temperature field and therefore the time-varying perturbation of the medium's optical response. Second, an optical forward model maps the SLM-modulated input field through this drifting medium. Third, a detector records the resulting output intensity distribution. Fourth, a feedback controller updates the SLM phase pattern in response to the measured output so that the desired intensity pattern can be maintained or recovered.

At the systems level, the operational chain is

$$E_{\mathrm{SLM}}(x, y) = E_{\mathrm{in}}(x, y)e^{i\phi(x,y)}, \qquad E_{\mathrm{out}} = T_{\mathrm{opt}}(t)E_{\mathrm{SLM}}, \qquad I(x, y) = |E_{\mathrm{out}}(x, y)|^2, \tag{1}$$

with $I(x, y)$ feeding the controller to generate the next $\phi(x, y)$. Here $E_{\mathrm{in}}(x, y)$ is the incident field at the SLM plane, $\phi(x, y)$ is the applied SLM phase pattern, $T_{\mathrm{opt}}(t)$ is the time-varying optical operator of the thermo-optical medium, and $I(x, y)$ is the measured output intensity distribution.

The thermal model and the controller are conceptually distinct. The thermal model—implemented as either a PINN-based surrogate or a classical numerical solver—predicts the evolving temperature field and, through the thermo-optic relation, the drift in $T_{\mathrm{opt}}(t)$. Its role is to describe how the medium changes in time, not to choose the SLM phase pattern. The controller closes the loop by acting on measured optical output, optionally augmented by model-based state estimates, to select the next $\phi(x, y)$. This controller may be reinforcement-learning-based, but it remains separate from the thermal model: the model explains the drift, while the controller compensates for it.

A schematic of the system is shown in Figure 1.

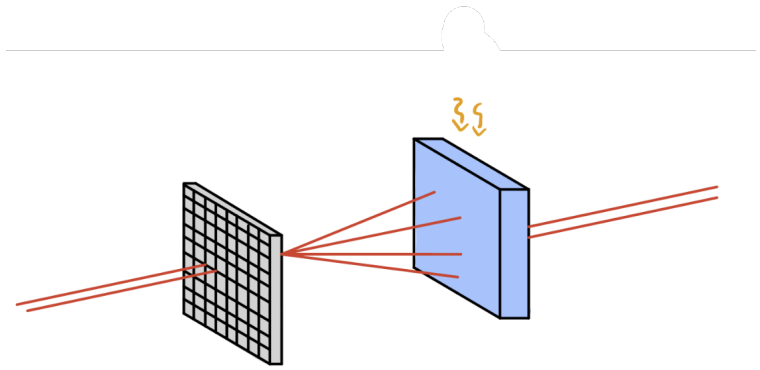

Figure 1: Schematic of the experimental setup. The laser beam is modulated by the SLM, passes through the thermo-optical material, and is detected by the camera.

The optical process is summarized as follows:

(I) Define the incident field $E_{\text{in}}(x, y)$ at the SLM plane.

(II) Apply SLM phase modulation to obtain

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y)e^{i\phi(x,y)}.$$

(III) Propagate through the thermo-optical material using the time-varying optical operator $T_{\text{opt}}(t)$ to obtain

$$E_{\text{out}} = T_{\text{opt}}(t)E_{\text{SLM}}.$$

(IV) Compute the output intensity

$$I(x, y) = |E_{\text{out}}(x, y)|^2.$$

(V) Use the measured output intensity to define the control objective, for example through a reward based on how well the observed output matches the desired target intensity pattern.

3 Background Information

3.1 Optical Physics

The optical beam is modelled as a Gaussian beam propagating along the optical axis (z-axis). Working in the paraxial approximation, the complex electric field is

$$E(x,y,z) = E_0 \frac{w_0}{w(z)} \exp\left(-\frac{x^2+y^2}{w(z)^2}\right) \exp\left(-ikz - i\frac{k(x^2+y^2)}{2R(z)} + i\zeta(z)\right), \tag{2}$$

where $k = 2\pi/\lambda$ is the wavenumber and the beam parameters are

$$w(z) = w_0\sqrt{1+\left(\frac{z}{z_R}\right)^2}, \tag{3}$$

$$R(z) = z\left(1+\left(\frac{z_R}{z}\right)^2\right), \tag{4}$$

$$\zeta(z) = \arctan\left(\frac{z}{z_R}\right), \qquad z_R = \frac{\pi w_0^2}{\lambda}. \tag{5}$$

Here w_0 is the beam waist, $w(z)$ is the beam radius at position z, $R(z)$ is the radius of curvature of the wavefronts, $\zeta(z)$ is the Gouy phase, and z_R is the Rayleigh range.

At the SLM plane $z = z_0$ the amplitude prefactor $w_0/w(z_0)$ is constant, and $\exp(-ikz_0)$ and $\exp(i\zeta(z_0))$ are *global* phases, so they may be dropped. The curvature term

$$\exp\left(-i\frac{k(x^2+y^2)}{2R(z_0)}\right)$$

is spatially varying and should be retained unless the beam is well collimated. The incident field at the SLM plane is therefore

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right)\exp\left(-i\frac{k(x^2+y^2)}{2R}\right), \tag{6}$$

where $w \equiv w(z_0)$, $R \equiv R(z_0)$, and A absorbs all constant prefactors. In the collimated limit ($R \to \infty$) this reduces to

$$E_{\text{in}}(x,y) = A\exp\left(-\frac{x^2+y^2}{w^2}\right). \tag{7}$$

The modulated field then propagates through thermo-optical material modelled by a linear transmission matrix $T_{\text{opt}}(t)$, and the detector measures only intensity:

$$E_{\text{out}} = T_{\text{opt}}(t)\, E_{\text{SLM}}, \qquad I(x,y) = |E_{\text{out}}(x,y)|^2. \tag{8}$$

Optical detectors measure intensity, not complex amplitude, so $I(x,y)$ is the only accessible quantity.

In summary, the incident field is the Gaussian beam evaluated at the SLM plane; spatially varying curvature phase is retained when needed; the downstream optical medium is captured by a complex transmission matrix; and the output camera measures intensity.

3.2 Thermal Physics

The optical medium is treated as a thermally perturbed phase/scattering element whose temperature field evolves on timescales much slower than the optical field. Over these slower thermal timescales, the temperature distribution $T(\mathbf{r}, t)$ is modelled by the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t), \tag{9}$$

where ρ is the density, c_p is the specific heat capacity, κ is the thermal conductivity, and $Q(\mathbf{r}, t)$ is any applied or absorbed heat source.

Temperature affects the optical response through the thermo-optic effect. To first order, the refractive index is taken to vary linearly with temperature:

$$n(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T(\mathbf{r}, t) - T_0), \tag{10}$$

where n_0 is the reference refractive index at temperature T_0, and dn/dT is the thermo-optic coefficient of the material. These refractive-index variations change the optical path length through the medium and therefore introduce a time-dependent phase perturbation

$$\Delta\phi(x, y, t) = k \int \Delta n(x, y, z, t)\, dz, \tag{11}$$

with $k = 2\pi/\lambda$ and $\Delta n = n - n_0$.

At the system level, these thermally induced phase changes are represented as a slow drift in the effective optical operator of the medium. In the transmission-matrix description used here, this is written as

$$T_{\text{opt}}(t) = T_{\text{opt},0} + \Delta T_{\text{opt}}(t), \tag{12}$$

where $T_{\text{opt},0}$ is the nominal transmission matrix and $\Delta T_{\text{opt}}(t)$ captures the time-varying perturbation caused by thermal diffusion. The key thermal-physics problem is therefore to model how heat input produces a temperature field, how that temperature field modifies refractive index, and how the resulting refractive-index changes alter the optical response seen by the detector.

3.3 SLM Behaviour

The SLM acts only in the transverse (x, y) plane and is modeled as a phase-only modulator. The applied phase pattern $\phi(x, y)$ is the control input and sets the relative phases across the aperture, thereby altering interference at the output. The modulated field is

$$E_{\text{SLM}}(x, y) = E_{\text{in}}(x, y) e^{i\phi(x,y)}.$$

3.4 Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) are a class of machine-learning models designed to approximate the solution of a physical system while explicitly respecting its governing equations. In a standard neural network, one typically provides input–output data pairs and trains the network to reproduce the observed mapping. A PINN extends this idea by requiring the network not only to fit data, but also to satisfy the differential equations, boundary conditions, and initial conditions that describe the underlying physics.

In the present project, the quantity of interest is the temperature field inside the thermo-optical material. The PINN represents this field by a feedforward neural network $T_\theta(\mathbf{r}, t)$, which takes a space-time point $(\mathbf{r}, t)$ as its input and produces a scalar temperature prediction as its output. Setting the input as the zeroth-layer activation, the forward pass proceeds layer by layer:

$$z^{[\ell]} = W^{[\ell]}\, a^{[\ell-1]} + b^{[\ell]}, \qquad a^{[\ell]} = \sigma\Big(z^{[\ell]}\Big), \tag{13}$$

for hidden layers $\ell = 1, \ldots, N-1$. Here $W^{[\ell]}$ is the weight matrix for layer ℓ, $b^{[\ell]}$ is the bias vector, $z^{[\ell]}$ is the pre-activation, and $a^{[\ell]}$ is the post-activation output. The final layer produces a single scalar with no output activation, interpreted directly as the predicted temperature $T_\theta(\mathbf{r}, t)$. The full set of trainable parameters is

$$\theta = \{W^{[1]}, b^{[1]}, \ldots, W^{[N]}, b^{[N]}\}, \tag{14}$$

and the role of training is to adjust θ so that the network output approximates a temperature field that is both physically admissible and consistent with any available observations.

For the thermal model considered here, the governing equation is the heat equation

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q(\mathbf{r}, t). \tag{15}$$

A PINN enforces this equation approximately by penalising violations at sampled collocation points. Substituting the network output into the left- and right-hand sides gives the PDE residual

$$\mathcal{R}_\theta(\mathbf{r}, t) = \rho c_p \frac{\partial T_\theta}{\partial t} - \nabla \cdot (\kappa \nabla T_\theta) - Q(\mathbf{r}, t). \tag{16}$$

If T_θ were the exact physical solution, this residual would vanish everywhere.

The overall training objective is a weighted sum of loss terms. Let $\{(\mathbf{r}_i^{(f)}, t_i^{(f)})\}_{i=1}^{N_f}$ denote interior collocation points used to enforce the PDE, $\{(\mathbf{r}_i^{(d)}, t_i^{(d)}, T_i^{\text{data}})\}_{i=1}^{N_d}$ the observed data points, $\{(\mathbf{r}_i^{(b)}, t_i^{(b)}, T_i^{\text{BC}})\}_{i=1}^{N_b}$ the boundary-condition samples, and $\{\mathbf{r}_i^{(0)}\}_{i=1}^{N_0}$ spatial points at the initial time $t = 0$ with prescribed initial temperature $T^{\text{init}}(\mathbf{r})$. The component losses are

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_f} \sum_{i=1}^{N_f} \Big| \mathcal{R}_\theta(\mathbf{r}_i^{(f)}, t_i^{(f)}) \Big|^2, \tag{17}$$

$$\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} \Big| T_\theta(\mathbf{r}_i^{(d)}, t_i^{(d)}) - T_i^{\text{data}} \Big|^2, \tag{18}$$

$$\mathcal{L}_{\text{BC}} = \frac{1}{N_b} \sum_{i=1}^{N_b} \Big| T_\theta(\mathbf{r}_i^{(b)}, t_i^{(b)}) - T_i^{\text{BC}} \Big|^2, \tag{19}$$

$$\mathcal{L}_{\text{IC}} = \frac{1}{N_0} \sum_{i=1}^{N_0} \Big| T_\theta(\mathbf{r}_i^{(0)}, 0) - T^{\text{init}}(\mathbf{r}_i^{(0)}) \Big|^2, \tag{20}$$

combined into the total loss

$$\mathcal{L} = \lambda_{\text{data}}\mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_{\text{IC}}\mathcal{L}_{\text{IC}}. \tag{21}$$

Here N_f, N_d, N_b, and N_0 are the sample counts for each constraint set, and T_i^{data}, T_i^{BC}, T^{init} are the prescribed temperature values at the corresponding sample points. The scalar coefficients λ_{data}, λ_{PDE}, λ_{BC}, and λ_{IC} are weighting hyperparameters that balance the relative contribution of each term during training.

The total loss $\mathcal{L}$ is a scalar function of θ; the partial derivatives of T_θ entering $\mathcal{R}_\theta$ are obtained by automatic differentiation through the network. Training then computes the gradients

$$\frac{\partial \mathcal{L}}{\partial W^{[\ell]}}, \qquad \frac{\partial \mathcal{L}}{\partial b^{[\ell]}} \tag{22}$$

and updates the weights and biases, for example by gradient descent:

$$W^{[\ell]} \leftarrow W^{[\ell]} - \eta \frac{\partial \mathcal{L}}{\partial W^{[\ell]}}, \qquad b^{[\ell]} \leftarrow b^{[\ell]} - \eta \frac{\partial \mathcal{L}}{\partial b^{[\ell]}}. \tag{23}$$

The intermediate quantities $z^{[\ell]}$ and $a^{[\ell]}$ are not updated directly; changing $W^{[\ell]}$ and $b^{[\ell]}$ changes the next forward pass and therefore $T_\theta(\mathbf{r}, t)$, which is how the physics constraints shape the learned solution.

Once a thermally consistent approximation $T_\theta(\mathbf{r}, t)$ has been obtained, it is coupled to the optical model through the thermo-optic relation

$$n_\theta(\mathbf{r}, t) = n_0 + \frac{dn}{dT}(T_\theta(\mathbf{r}, t) - T_0), \tag{24}$$

which gives a corresponding refractive-index field. This induces a time-dependent optical phase perturbation,

$$\Delta\phi_\theta(x, y, t) = k \int (n_\theta(x, y, z, t) - n_0)\, dz, \tag{25}$$

and therefore modifies the effective optical response of the material. In the present setting, the PINN is thus used as a physics-constrained surrogate model for the thermal dynamics that drive slow drift in the optical system.

The main appeal of this approach is that it incorporates known thermal physics directly into the learning process, reducing reliance on large labelled datasets and yielding predictions that remain consistent with the governing equations even when the internal state of the medium is only partially observed through optical measurements.

3.5 Reinforcement Learning

Reinforcement learning (RL) is used here as the feedback-control layer that updates the SLM phase pattern in response to measured output intensity. The environment is the optical system with its thermally drifting medium; at each time step t the agent observes s_t derived from $I(x, y)$ (optionally augmented by the thermal model), applies an action a_t corresponding to a phase pattern $\phi(x, y)$ or its update, and receives a reward such as

$$r_t = I(x_0, y_0, t), \tag{26}$$

the intensity at a designated target location. The learned policy $\pi(a_t \mid s_t)$ therefore encodes a phase-update strategy that maintains target intensity in the presence of thermal drift.

The RL controller is distinct from the thermal model. The PINN or numerical solver predicts the evolving temperature field and thus the drift in $T_{\mathrm{opt}}(t)$; that information may be used to shape s_t or to build a training simulator, but the control loop itself is driven by optical measurements. This separation clarifies how modelling and control couple in the overall task: the thermal model tracks drift, while RL compensates it via SLM phase updates.